Bug Reports — HW5: Recovery / Fault Injection

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Defects found by injecting faults: sleep, watchdog (WDT) hang, interrupted save and sensor loss. Release decision in the source report: NO-GO.

Environment: ESP32 DevKit V1: DHT22 on GPIO15, button on GPIO4, LED on GPIO2, relay on GPIO5; Wokwi

Summary

ID	Title	Severity
Bug_1	System wakes in less than 1 s after the "sleep 5" command	Major
Bug_3	Relay is ON after the "hang" command	Critical
Bug_4	The "sensor" command returns data when the sensor is disconnected	Critical

Note: Bug_2 ("WDT reset in ~30 s instead of ≤5 s") is not included: it was withdrawn after mentor feedback. The 30 s was real wall-clock time, because Wokwi slows down to ≈2% speed after "hang". By the Wokwi simulation timer, [FAULT] was at ≈11:52.4 and the reboot at ≈11:55.4, i.e. ≈3 s, which meets the requirement. Original numbering (Bug_1, Bug_3, Bug_4) is kept.

Bug_1 — System wakes in less than 1 s after the "sleep 5" command

Requirement	The `sleep <N>` command puts the device into deep sleep for approximately N seconds.
Risk	Wrong state timing
Preconditions	Device ACTIVE
Steps	1. Enter the "sleep 5" command. 2. Check the time when the system wakes.
Expected	System wakes in 4.25–5.75 s
Actual	System wakes in less than 1 s
Timing / Reset reason	deep sleep
Evidence	Serial log; video: https://drive.google.com/file/d/1sDcnInEVqLWMa99T-0tNufTBgGRBEwIy/view?usp=sharing
Reproducibility	5/5
Severity	Major

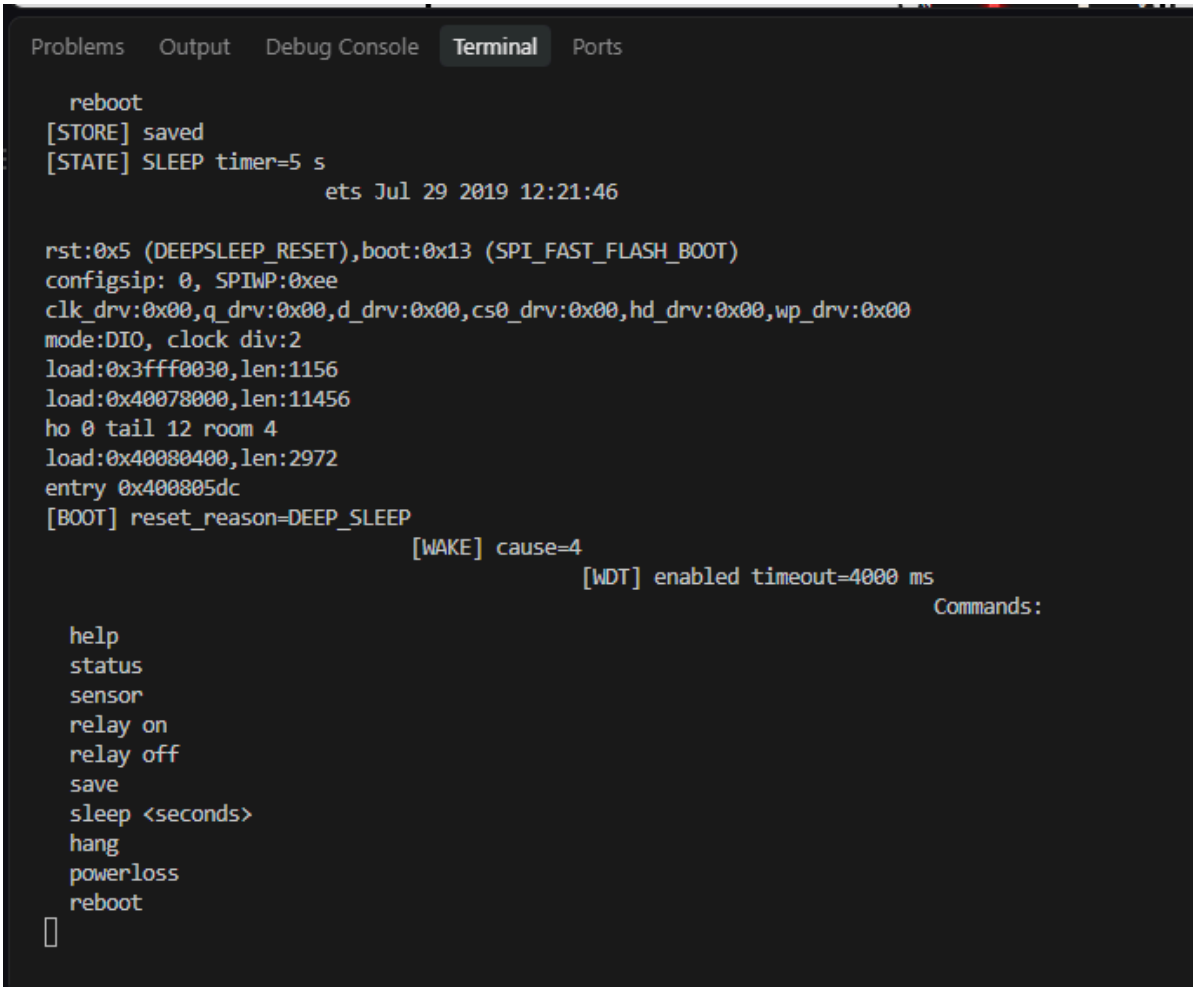


Figure: Sleep timing (serial log)

Bug_3 — Relay is ON after the "hang" command

Requirement	The actuator must be OFF after an emergency reset.
Risk	Unsafe recovery
Preconditions	Device ACTIVE, relay ON
Steps	1. Enter the "hang" command. 2. Enter the "status" command. 3. Check the log.
Expected	After the WDT reset the relay is OFF
Actual	After the WDT reset the relay is ON
Timing / Reset reason	INT_WDT
Evidence	Serial log
Reproducibility	5/5
Severity	Critical

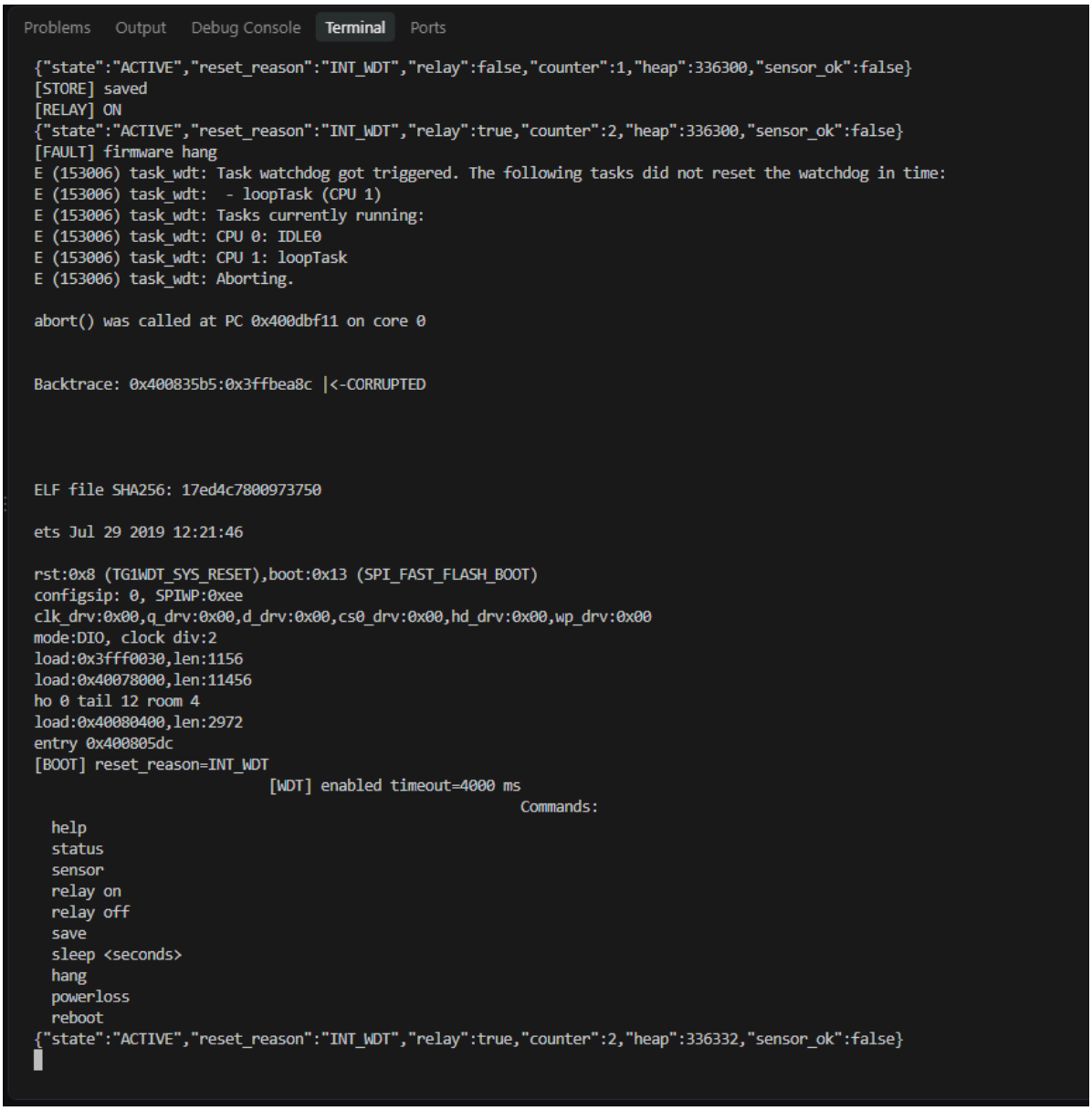


Figure: Relay + WDT (serial log)

Bug_4 — The "sensor" command returns data when the sensor is disconnected

Requirement	If the DHT22 is unavailable or returns incorrect data: the firmware must not treat it as a valid measurement; there must be an explicit error in the logs.
Risk	Invalid sensor data
Preconditions	Device ACTIVE
Steps	1. Disconnect the DHT22. 2. Enter the "sensor" command. 3. Check the log.
Expected	Explicit sensor error
Actual	[SENSOR] temp=0.0C humidity=0.0%
Timing / Reset reason	POWER_ON
Evidence	Serial log

Reproducibility	5/5
Severity	Critical

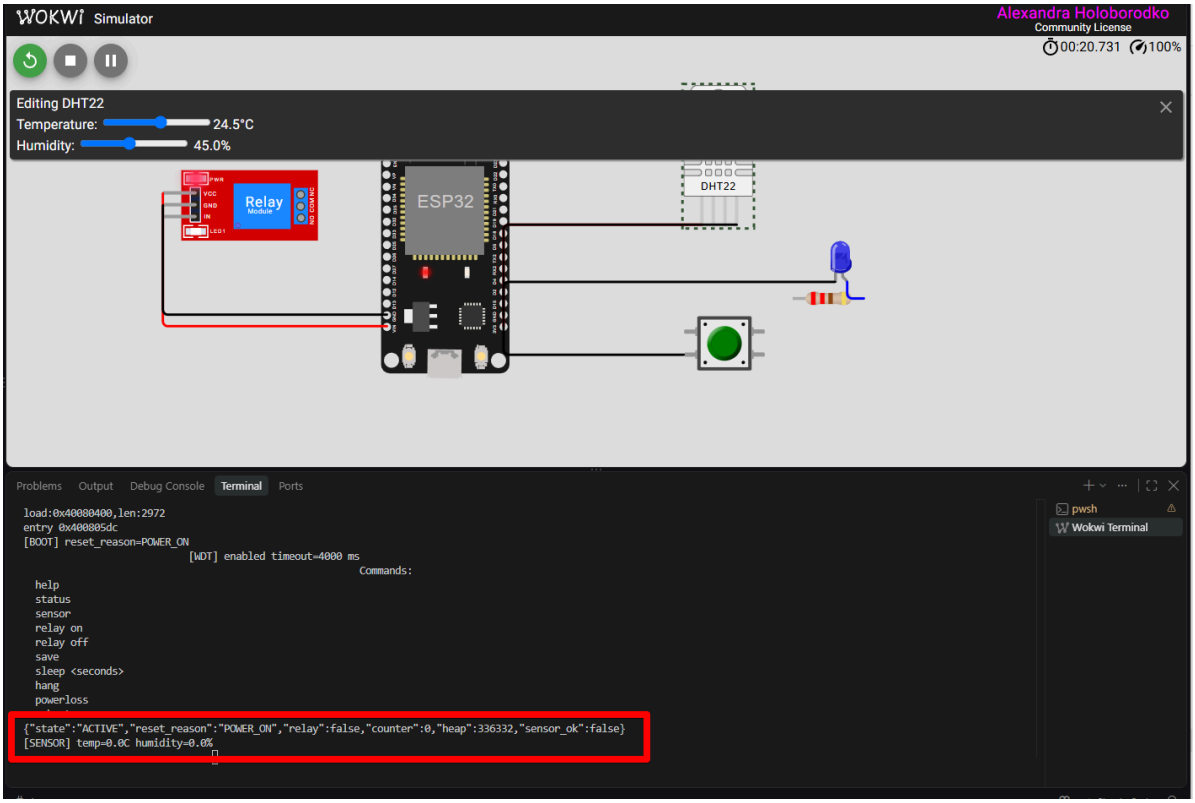


Figure: Sensor unavailable